

Quantum-Chemistry Approach to Multi-Species Ultra-Cold Atoms in Optical Tweezer Arrays

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March 20, 2026

History

History

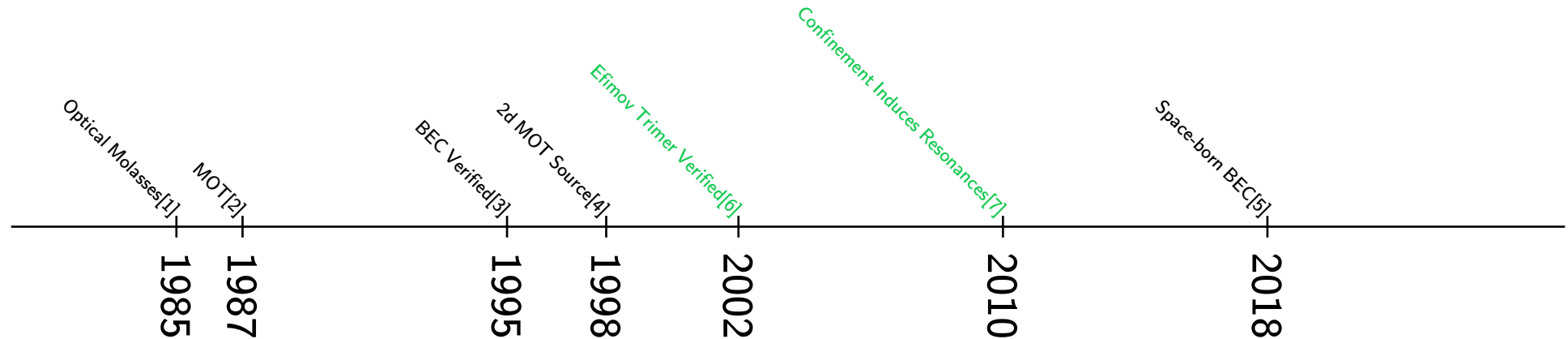
black : General, red : Quantum Simulator, blue : Quantum Computer, green : Few Body Physics purple : Numerics



- [1] S. Chu *et al.*, "Three-dimensional viscous confinement and cooling of atoms by resonance radiation pressure," *Phys. Rev. Lett.*, 1985.
- [2] E. L. e. a. Raab, "Trapping of Neutral Sodium Atoms with Radiation Pressure," *Phys. Rev. Lett.*, 1987.
- [3] M. H. A. el al., "Observation of Bose-Einstein condensation in a dilute atomic vapor," *Science*, 1995.
- [4] K. Dieckmann *et al.*, "Two-dimensional magneto-optical trap as a source of slow atoms," *Phys. Rev. A*, 1998.
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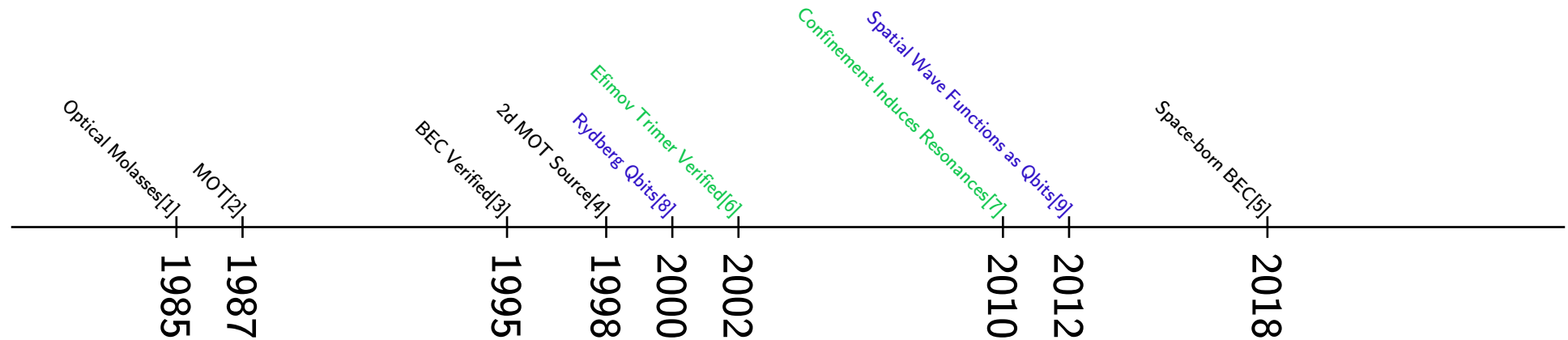


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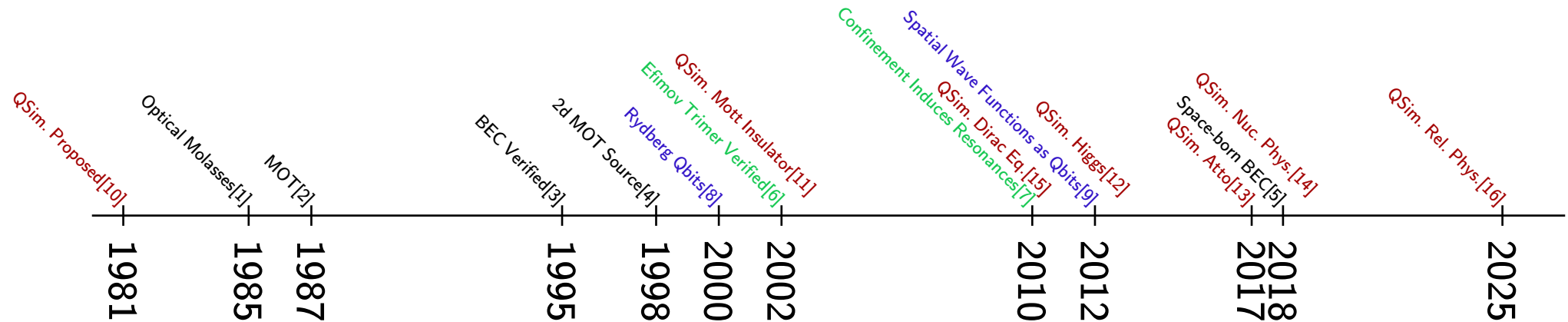


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[9] P.-I. Schneider *et al.*, "Quantum computation with ultracold atoms in a driven optical lattice," *Phys. Rev. A*, 2012.

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[11] I. Bloch *et al.*, "Quantum phase transition from a superfluid to a Mott insulator in a gas of ultracold atoms," *Nature*, 2002.

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[13] S. Sala *et al.*, "Ultracold-atom quantum simulator for attosecond science." p. 11403, 2017.

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[15] C. F. Roos *et al.*, "Quantum simulation of the Dirac equation," *Nature*, 2010.

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Quantum Simulator

- Optical Lattices, Orthogonal (reflected) Laser Beams

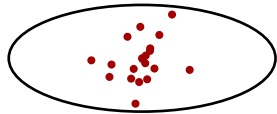
- Optical Lattices, Orthogonal (reflected) Laser Beams
- Optical Tweezer Arrays, Highly focused Laser Beam

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- Optical Tweezer Arrays, Highly focused Laser Beam
- Single-site manipulation

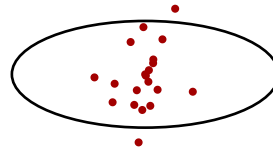
- Optical Lattices, Orthogonal (reflected) Laser Beams
- Optical Tweezer Arrays, Highly focused Laser Beam
- Single-site manipulation
- Feshbach Resonances → Tunable (effective) interaction-strength

Quantum Simulator: Hubbard-Modell (2002)

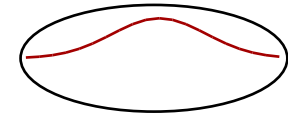
$$H = -J \sum_{\langle i,j \rangle} a_i^\dagger a_j + \sum_i e_i n_i + \frac{1}{2} U \sum_i n_i (n_i - 1)$$



Step 1: Load atoms into cigar shaped trap



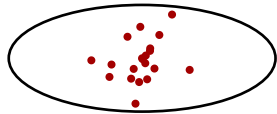
Step 2: Evaporative cooling



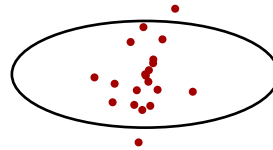
Step 3: BEC Formation

Quantum Simulator: Hubbard-Modell (2002)

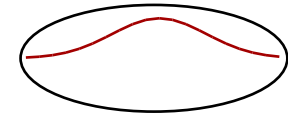
$$H = -J \sum_{\langle i,j \rangle} a_i^\dagger a_j + \sum_i e_i n_i + \frac{1}{2} U \sum_i n_i (n_i - 1)$$



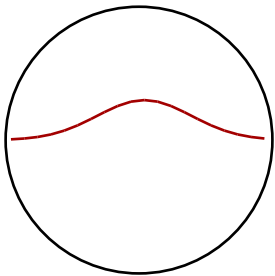
Step 1: Load atoms into cigar shaped trap



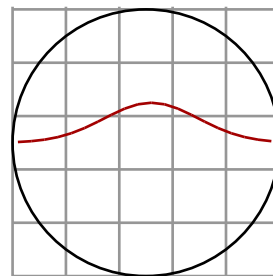
Step 2: Evaporative cooling



Step 3: BEC Formation



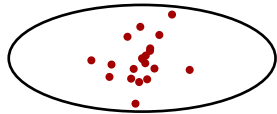
Step 4: Relax trap to isotropic trap



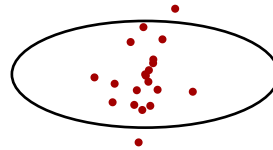
Step 5: Add optical lattice

Quantum Simulator: Hubbard-Modell (2002)

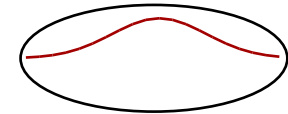
$$H = -J \sum_{\langle i,j \rangle} a_i^\dagger a_j + \sum_i e_i n_i + \frac{1}{2} U \sum_i n_i (n_i - 1)$$



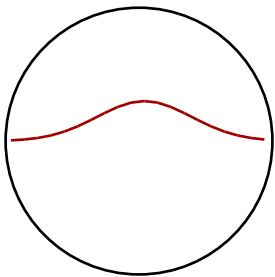
Step 1: Load atoms into cigar shaped trap



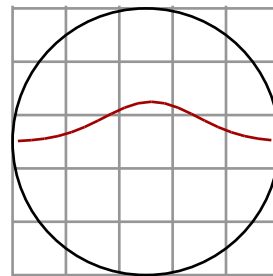
Step 2: Evaporative cooling



Step 3: BEC Formation



Step 4: Relax trap to isotropic trap



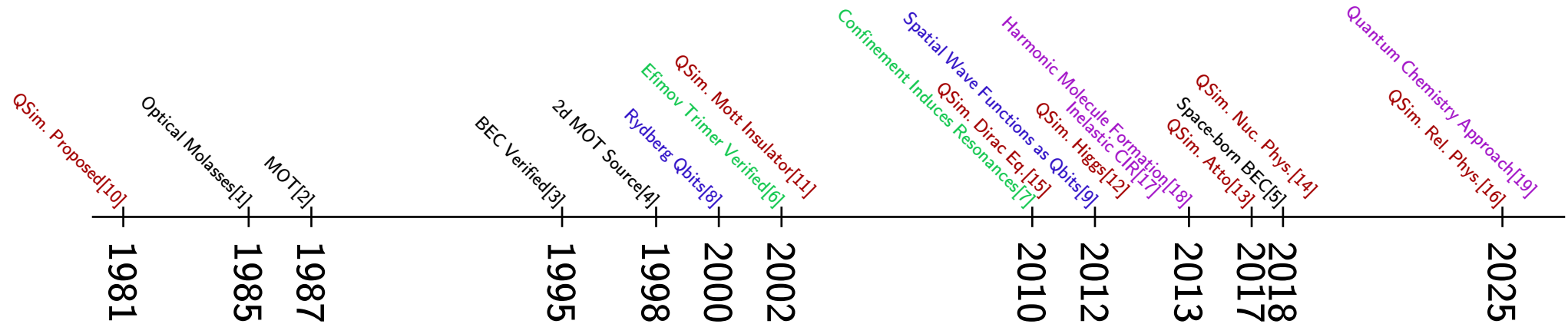
Step 5: Add optical lattice



Step 6: Analyse interference pattern

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[17] S. Sala *et al.*, “Inelastic Confinement-Induced Resonances in Low-Dimensional Quantum Systems,” *Physical Review Letter*, 2012.

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[19] M. u. Rehman *et al.*, “Confined few-particle systems beyond mean-field theory adopting Gaussian-type orbitals and Morse interparticle interaction,” *Physical Review A*, vol. 112, p. 63320, 2025.

Quantum Chemistry Approach

Ultra-Cold Atoms

Quantum-Chemistry

Ultra-Cold Atoms

- Few Body

Quantum-Chemistry

- Few Body

Ultra-Cold Atoms

- Few Body
- Non-relativistic $\Rightarrow$ Schrödinger Equation

Quantum-Chemistry

- Few Body
- Non-relativistic $\Rightarrow$ Schrödinger Equation

Ultra-Cold Atoms

- Few Body
- Non-relativistic $\Rightarrow$ Schrödinger Equation
- Variable Geometry

Quantum-Chemistry

- Few Body
- Non-relativistic $\Rightarrow$ Schrödinger Equation
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Ultra-Cold Atoms

- Few Body
- Non-relativistic $\Rightarrow$ Schrödinger Equation
- Variable Geometry
- **Inter-atomic Potential**

Quantum-Chemistry

- Few Body
- Non-relativistic $\Rightarrow$ Schrödinger Equation
- Variable Geometry
- **Coulomb Interaction**

Ultra-Cold Atoms

- Few Body
- Non-relativistic $\Rightarrow$ Schrödinger Equation
- Variable Geometry
- Inter-atomic Potential
- Bosons and Fermions

Quantum-Chemistry

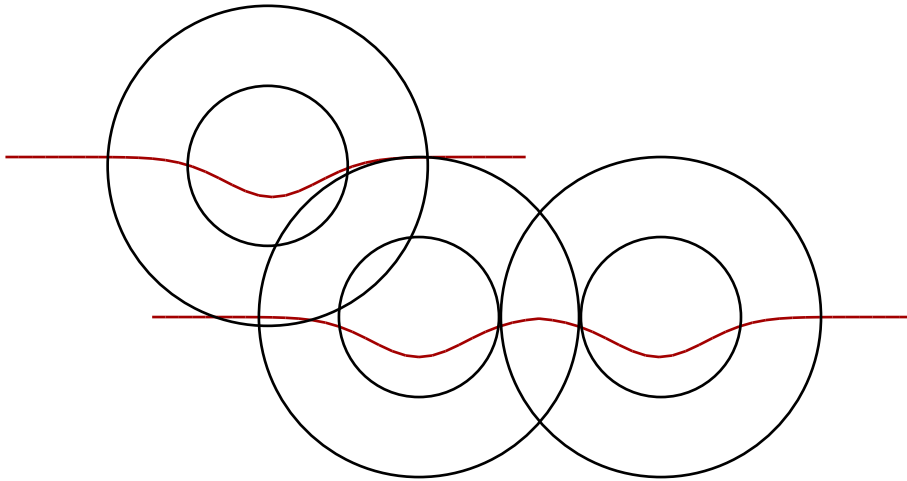
- Few Body
- Non-relativistic $\Rightarrow$ Schrödinger Equation
- Variable Geometry
- Coulomb Interaction
- Indistinguishable Fermions

$$\varphi_{\text{gto}}^{\sigma, R, \alpha} = N \prod_{j=1}^3 (x_j - R_j)^{\sigma_j} \exp\left(-\alpha (x_j - R_j)^2\right)$$

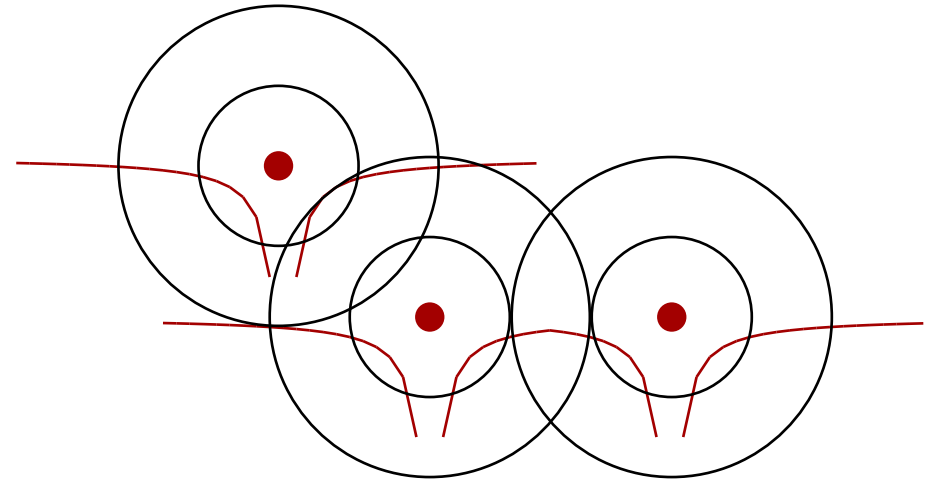
Quantum-Chemistry Approach: Cartesian Gaussian Type Orbitals (CGTOs)

$$\varphi_{\text{gto}}^{\sigma, R, \alpha} = N \prod_{j=1}^3 (x_j - R_j)^{\sigma_j} \exp\left(-\alpha(x_j - R_j)^2\right)$$

Ultra-Cold Atoms (red: traps)



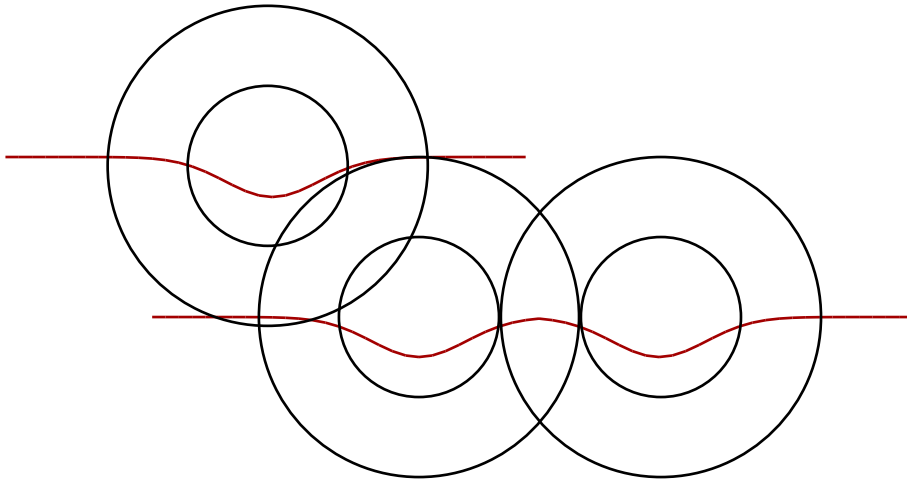
Quantum-Chemistry (red: cores)



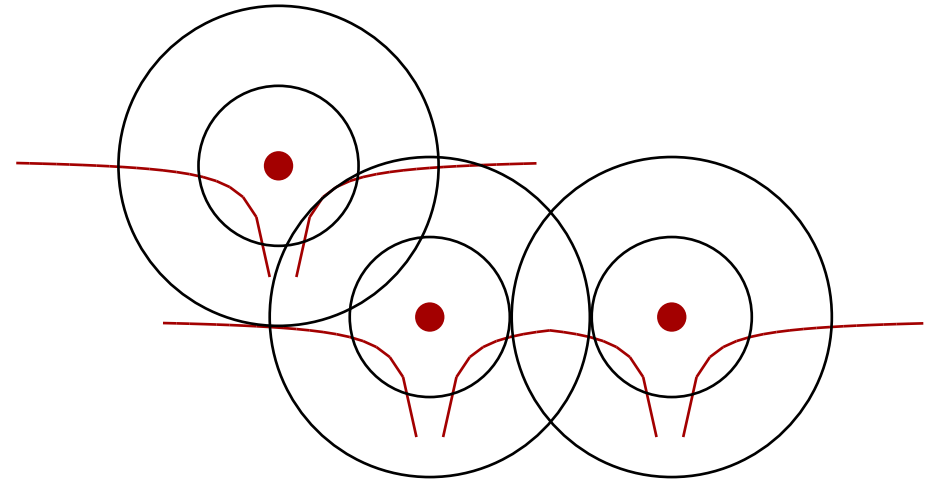
Quantum-Chemistry Approach: Cartesian Gaussian Type Orbitals (CGTOs)

$$\varphi_{\text{gto}}^{\sigma, R, \alpha} = N \prod_{j=1}^3 (x_j - R_j)^{\sigma_j} \exp\left(-\alpha(x_j - R_j)^2\right)$$

Ultra-Cold Atoms (red: traps)



Quantum-Chemistry (red: cores)

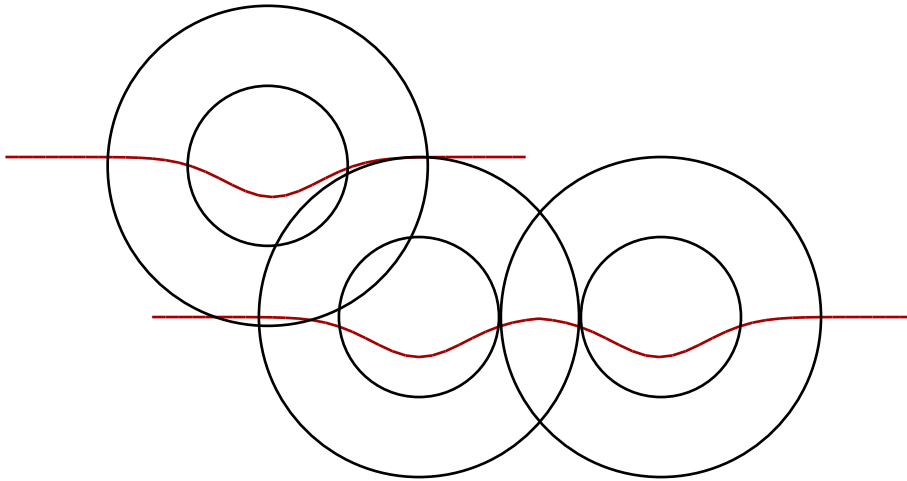


Traps $\leftrightarrow$ Cores
Atoms $\leftrightarrow$ Electrons

Quantum-Chemistry Approach: Cartesian Gaussian Type Orbitals (CGTOs)

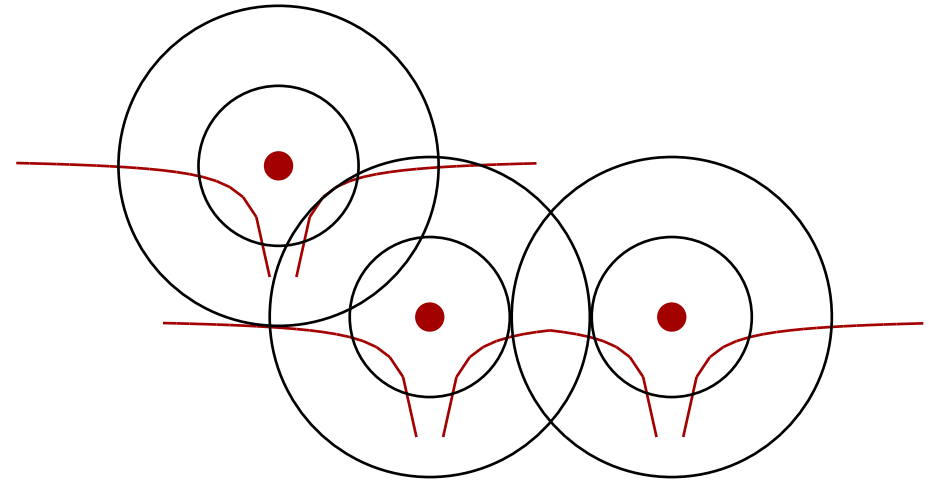
$$\varphi_{\text{gto}}^{\sigma, R, \alpha} = N \prod_{j=1}^3 (x_j - R_j)^{\sigma_j} \exp\left(-\alpha(x_j - R_j)^2\right)$$

Ultra-Cold Atoms (red: traps)



CGTOs are phys. motivated

Quantum-Chemistry (red: cores)



CGTOs are **not** phys. motivated

Traps $\leftrightarrow$ Cores
Atoms $\leftrightarrow$ Electrons

Multi-Species Self-Consistent-Field Theory (MSSCF)

- Use product wave ansatz

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- Construct energy functional $E[\Psi] = \langle \Psi | H | \Psi \rangle$

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- Minimize energy functional w.r.t single-particle wave function

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- Coupled Fock equation for every species

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- Construct energy functional $E[\Psi] = \langle \Psi | H | \Psi \rangle$
- Minimize energy functional w.r.t single-particle wave function
- Coupled Fock equation for every species
- (Extended) Pople-Nesbet equation

$$H = \sum_i^{\mathcal{N}_B} \sum_{n=1}^{N_i} h_i(r_{i,n}) + \sum_i^{\mathcal{N}_F} \sum_{n=1}^{N_i} h_i(r_{i,n})$$

$\mathcal{N}_B, \mathcal{N}_F$: number of bosonic and fermionic species N_i : number of particles in i th species

$$H = \sum_i^{\mathcal{N}_B} \sum_{n=1}^{N_i} h_i(r_{i,n}) + \sum_i^{\mathcal{N}_F} \sum_{n=1}^{N_i} h_i(r_{i,n}) \\ - \frac{1}{2} \sum_i^{\mathcal{N}_B} \sum_{n \neq m}^{N_i} w^{(ii)}(r_{i,n}, r_{i,m}) - \frac{1}{2} \sum_i^{\mathcal{N}_F} \sum_{n \neq m}^{N_i} w^{(ii)}(r_{i,n}, r_{i,m})$$

$\mathcal{N}_B, \mathcal{N}_F$: number of bosonic and fermionic species N_i : number of particles in i th species

$$\begin{aligned} H = & \sum_i^{\mathcal{N}_B} \sum_{n=1}^{N_i} h_i(r_{i,n}) + \sum_i^{\mathcal{N}_F} \sum_{n=1}^{N_i} h_i(r_{i,n}) \\ & - \frac{1}{2} \sum_i^{\mathcal{N}_B} \sum_{n \neq m}^{N_i} w^{(ii)}(r_{i,n}, r_{i,m}) - \frac{1}{2} \sum_i^{\mathcal{N}_F} \sum_{n \neq m}^{N_i} w^{(ii)}(r_{i,n}, r_{i,m}) \\ & - \frac{1}{2} \sum_{i,j}^{\mathcal{N}_B} \sum_{n,m}^{N_i, N_j} w^{(ij)}(r_{i,n}, r_{j,m}) - \frac{1}{2} \sum_{i,j}^{\mathcal{N}_F} \sum_{n,m}^{N_i, N_j} w^{(ij)}(r_{i,n}, r_{j,m}) \end{aligned}$$

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$$\begin{aligned}
 H = & \sum_i^{\mathcal{N}_B} \sum_{n=1}^{N_i} h_i(r_{i,n}) + \sum_i^{\mathcal{N}_F} \sum_{n=1}^{N_i} h_i(r_{i,n}) \\
 & - \frac{1}{2} \sum_i^{\mathcal{N}_B} \sum_{n \neq m}^{N_i} w^{(ii)}(r_{i,n}, r_{i,m}) - \frac{1}{2} \sum_i^{\mathcal{N}_F} \sum_{n \neq m}^{N_i} w^{(ii)}(r_{i,n}, r_{i,m}) \\
 & - \frac{1}{2} \sum_{i,j}^{\mathcal{N}_B} \sum_{n,m}^{N_i, N_j} w^{(ij)}(r_{i,n}, r_{j,m}) - \frac{1}{2} \sum_{i,j}^{\mathcal{N}_F} \sum_{n,m}^{N_i, N_j} w^{(ij)}(r_{i,n}, r_{j,m}) \\
 & - \sum_i^{\mathcal{N}_B} \sum_j^{\mathcal{N}_F} \sum_{n,m}^{N_i, N_j} w^{(ij)}(r_{i,n}, r_{j,m})
 \end{aligned}$$

$\mathcal{N}_B, \mathcal{N}_F$: number of bosonic and fermionic species N_i : number of particles in i th species

- Fermions: Well known Hartree-Fock ansatz

$$|\mathcal{F}^i\rangle = \sum_{\sigma \in \mathcal{S}_N} (-1)^{\# \sigma} \varphi_{\sigma(1)}(r_1) \dots \varphi_{\sigma(N)}(r_N)$$

- Fermions: Well known Hartree-Fock ansatz

$$|\mathcal{F}^i\rangle = \sum_{\sigma \in \mathcal{S}_N} (-1)^{\# \sigma} \varphi_{\sigma(1)}(r_1) \dots \varphi_{\sigma(N)}(r_N)$$

- Bosons: All Bosons are in ground-state

$$|\mathcal{B}^i\rangle = \varphi_1(r_1) \varphi_1(r_2) \dots \varphi_1(r_N)$$

Bosonic Species

$$\begin{aligned}\mathbb{F}_{\text{bos}}^{\kappa}(c^1, \dots, c^{\mathcal{N}}) &= \mathbb{h}_{km}^{\kappa} + (N_{\kappa} - 1) \sum_{l,n} w_{klmn}^{\kappa\kappa} c_l^{\kappa} c_n^{\kappa} \\ &+ \sum_{i \neq \kappa}^{\mathcal{N}_B} N_i \sum_{l,n} w_{klmn}^{i\kappa} c_l^i c_n^i \\ &+ \sum_{i=\mathcal{N}_B+1}^{\mathcal{N}} \sum_{z=1}^{N_j} \sum_{l,n} w_{klmn}^{j\kappa} c_l^{j,z} c_n^{j,z}\end{aligned}$$

Fermionic Species

$$\begin{aligned}\mathbb{F}_{\text{fer}}^{\kappa}(c^1, \dots, c^{\mathcal{N}}) &= \mathbb{h}_{km}^{\kappa} + \sum_{l,n} (w_{klmn}^{\kappa\kappa} - w_{klnm}^{\kappa\kappa}) c_l^{\kappa,z} c_n^{\kappa,z} \\ &+ \sum_{i=\mathcal{N}_B+1}^{\mathcal{N}} \sum_{z=1}^{N_i} \sum_{l,n} w_{klmn}^{i\kappa} c_l^{i,z} c_n^{i,z} \\ &+ \sum_{i=1}^{\mathcal{N}_b} N_i \sum_{l,n} w_{klmn}^{i\kappa} c_l^i c_n^i\end{aligned}$$

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$$\mathbb{F}_{\text{bos}}^{\kappa}(c^1 \dots, c^{\mathcal{N}}) c^{\kappa} = \varepsilon c^{\kappa}$$

Fermionic Species

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$$\mathbb{F}_{\text{fer}}^{\kappa}(c^1 \dots, c^{\mathcal{N}}) c^{\kappa} = \varepsilon c^{\kappa}$$

Multi-Species Full Configuration Interaction (MSFCI)

Bosonic Species

$$|\mathcal{B}^i\rangle = \sum_{\sigma \in \mathcal{S}_N} (+1)^{\# \sigma} \varphi_{\sigma(1)}(r_1) \cdots \varphi_{\sigma(N)}(r_N)$$

Fermionic Species

$$|\mathcal{F}^i\rangle = \sum_{\sigma \in \mathcal{S}_N} (-1)^{\# \sigma} \varphi_{\sigma(1)}(r_1) \cdots \varphi_{\sigma(N)}(r_N)$$

Bosonic Species

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No Pauli exclusion principle

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Pauli exclusion principle

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No Pauli exclusion principle

Draw **with** repetition $\binom{M+N_i-1}{N_i}$

Fermionic Species

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Pauli exclusion principle

Draw **without** repetition $\binom{M}{N_i}$

Bosonic Species

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No Pauli exclusion principle

Draw **with** repetition $\binom{M+N_i-1}{N_i}$

Encoding e.g. 000, 001, 555

Fermionic Species

$$|\mathcal{F}^i\rangle = \sum_{\sigma \in \mathcal{S}_N} (-1)^{\# \sigma} \varphi_{\sigma(1)}(r_1) \cdots \varphi_{\sigma(N)}(r_N)$$

Pauli exclusion principle

Draw **without** repetition $\binom{M}{N_i}$

Encoding e.g. 123, 124, 345

Bosonic Species

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Non-trivial point group $\mathcal{G}$

Fermionic Species

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Draw **without** repetition $\binom{M}{N_i}$

Encoding e.g. 123, 124, 345

Trivial point group

$$|\mathcal{B}^i\rangle = \sum_{\sigma \in \frac{\mathcal{S}_n}{\mathcal{G}}} \sum_{\rho \in \mathcal{G}} \varphi((\sigma.\rho)(1)) \dots \varphi((\sigma.\rho)(N_i)) , \quad \mathfrak{B} = \prod_i^{\mathcal{N}} \mathfrak{B}^i , \quad \text{Fermions} \subset \text{Bosons}$$

MSFCI: Algorithm

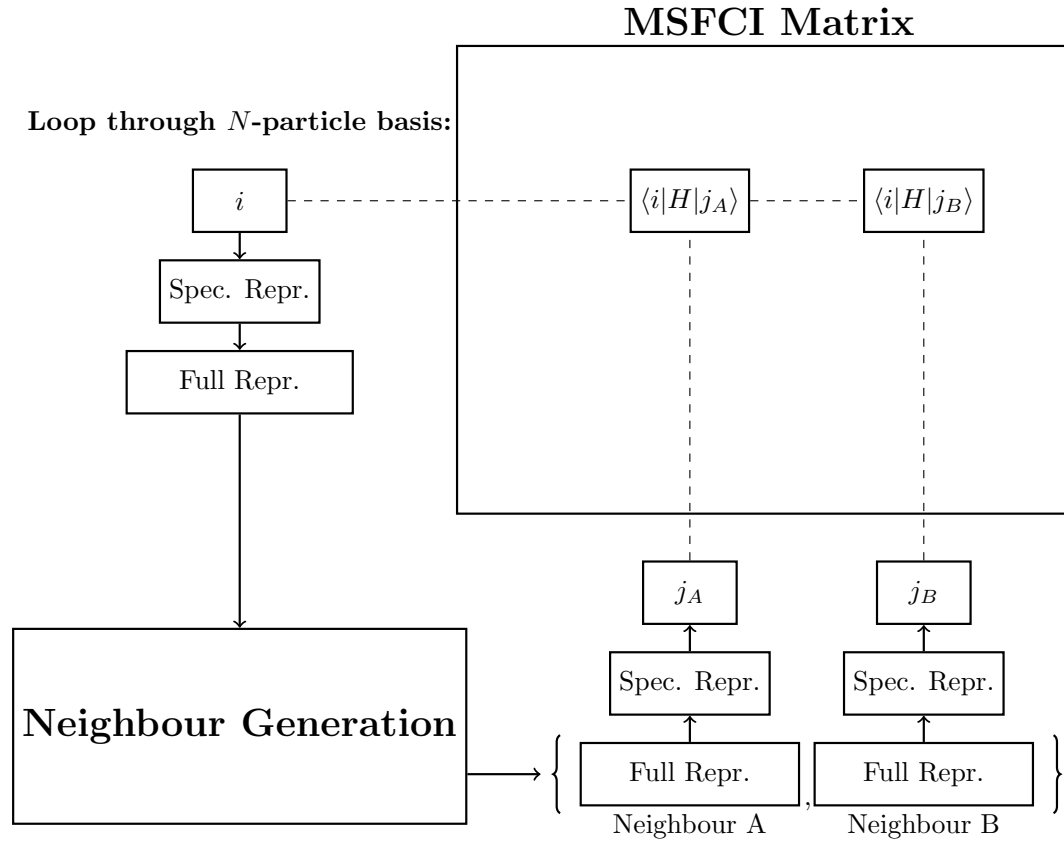


Figure 1: “MSFCI equation matrix”

Multi-Species Complete Active Space Configuration Interaction (MSCASCI)

- Separate space: Core, Active, Virtual

- Separate space: Core, Active, Virtual
- Core Orbitals : Occupied

- Separate space: Core, Active, Virtual
- Core Orbitals : Occupied
- Virtual Orbitals: Unoccupied

- Separate space: Core, Active, Virtual
- Core Orbitals : Occupied
- Virtual Orbitals: Unoccupied
- Active Orbitals: **Complete CI**

- Separate space: Core, Active, Virtual
- Core Orbitals : Occupied
- Virtual Orbitals: Unoccupied
- Active Orbitals: **Complete CI**
- Reduced CI space $\Rightarrow$ Drastically improved performance

- Separate space: Core, Active, Virtual
- Core Orbitals : Occupied
- Virtual Orbitals: Unoccupied
- Active Orbitals: **Complete CI**
- Reduced CI space $\Rightarrow$ Drastically improved performance
- CASCI Basis $\leftrightarrow$ FCI Basis

- Separate space: Core, Active, Virtual
- Core Orbitals : Occupied
- Virtual Orbitals: Unoccupied
- Active Orbitals: **Complete CI**
- Reduced CI space $\Rightarrow$ Drastically improved performance
- CASCI Basis $\leftrightarrow$ FCI Basis

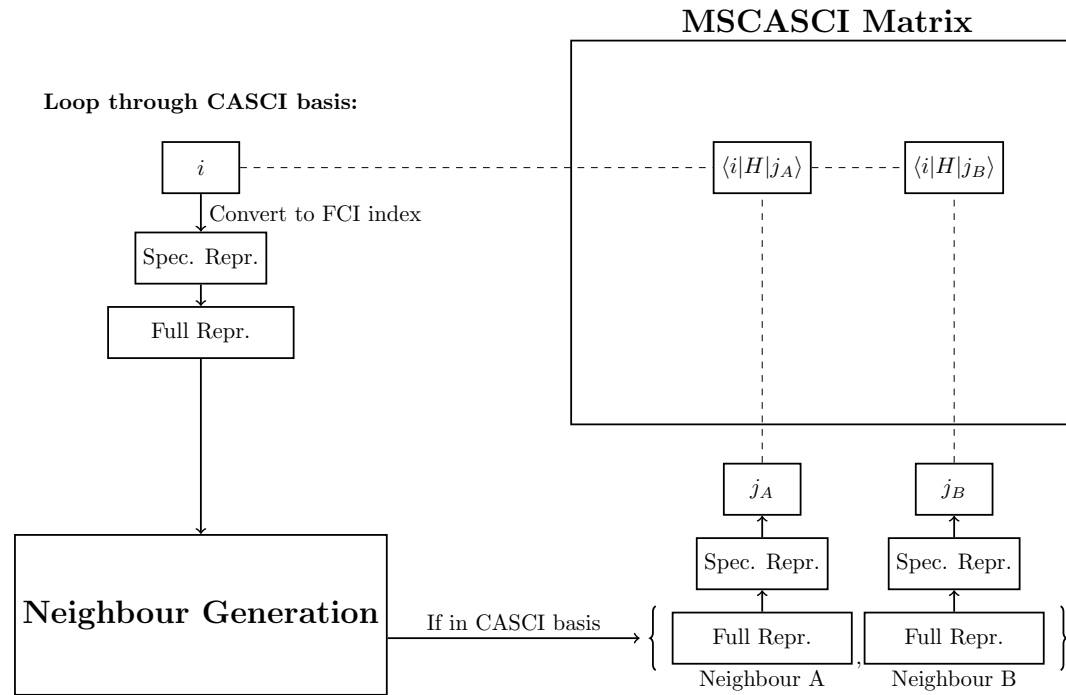


Figure 2: "MSCASCI as a modified MSFCI algorithm"

Verification & Preliminary Results

Verification and Results: 2 Boson MSFCI

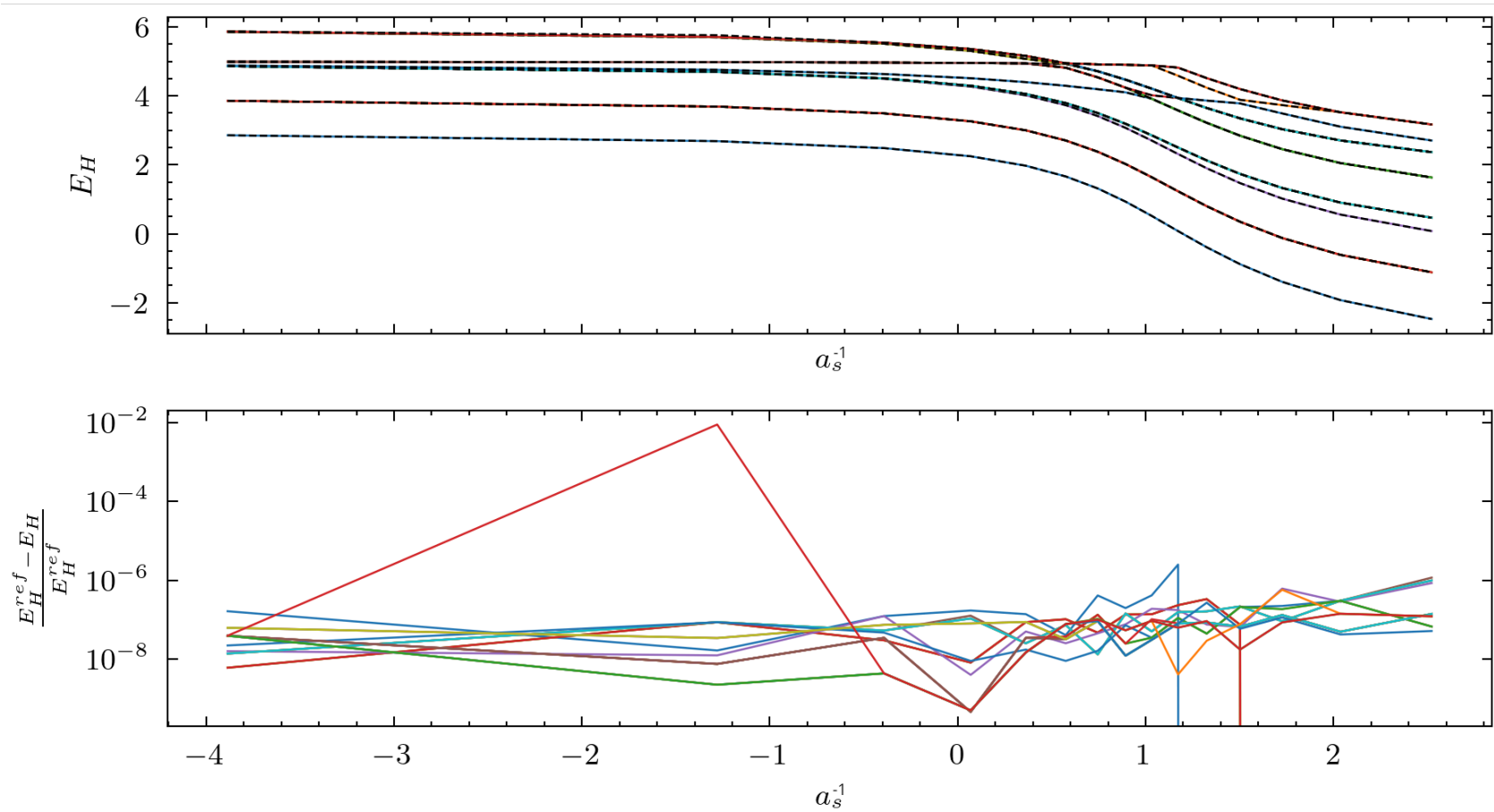


Figure 3: 2 Boson data (from [19] M. u. Rehman *et al.*, “Confined few-particle systems beyond mean-field theory adopting Gaussian-type orbitals and Morse interparticle interaction,” *Physical Review A*, vol. 112, p. 63320, 2025.) reproduced. **Solid: Literature data. Dashed: MSFCI data**

Verification and Results: MSSCF / MSFCI Comparison

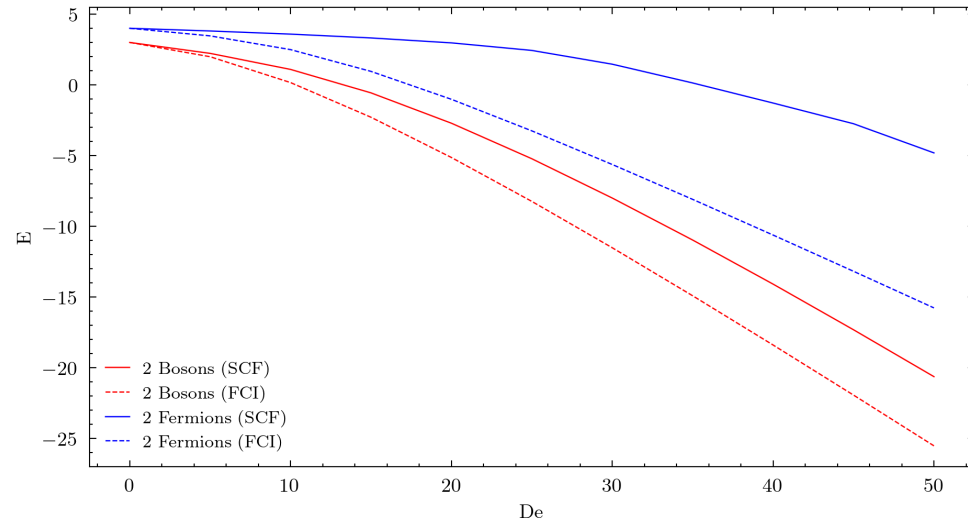


Figure 4: 2 Body MSSCF / MSFCI Comparison

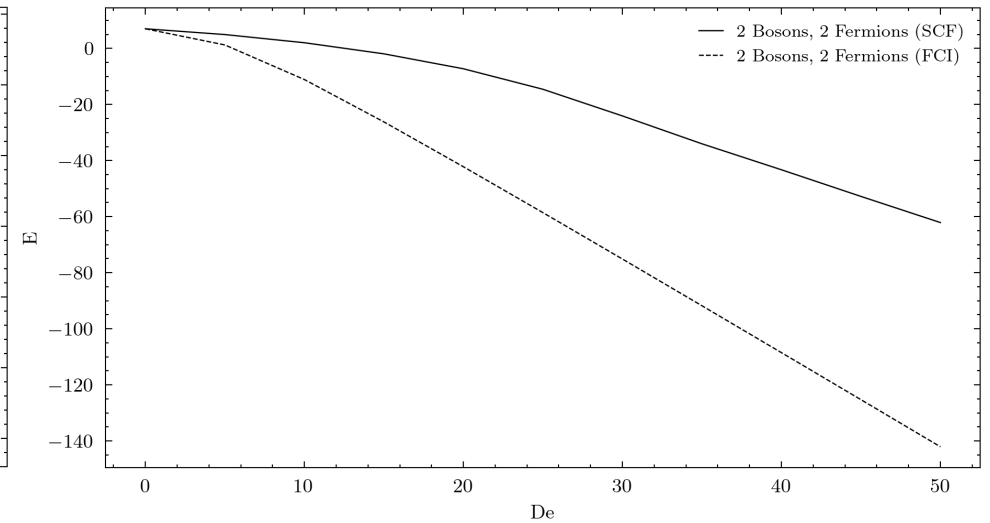


Figure 5: 4 Body MSSCF / MSFCI Comparison

Verification and Results: 3 Boson MSFCI

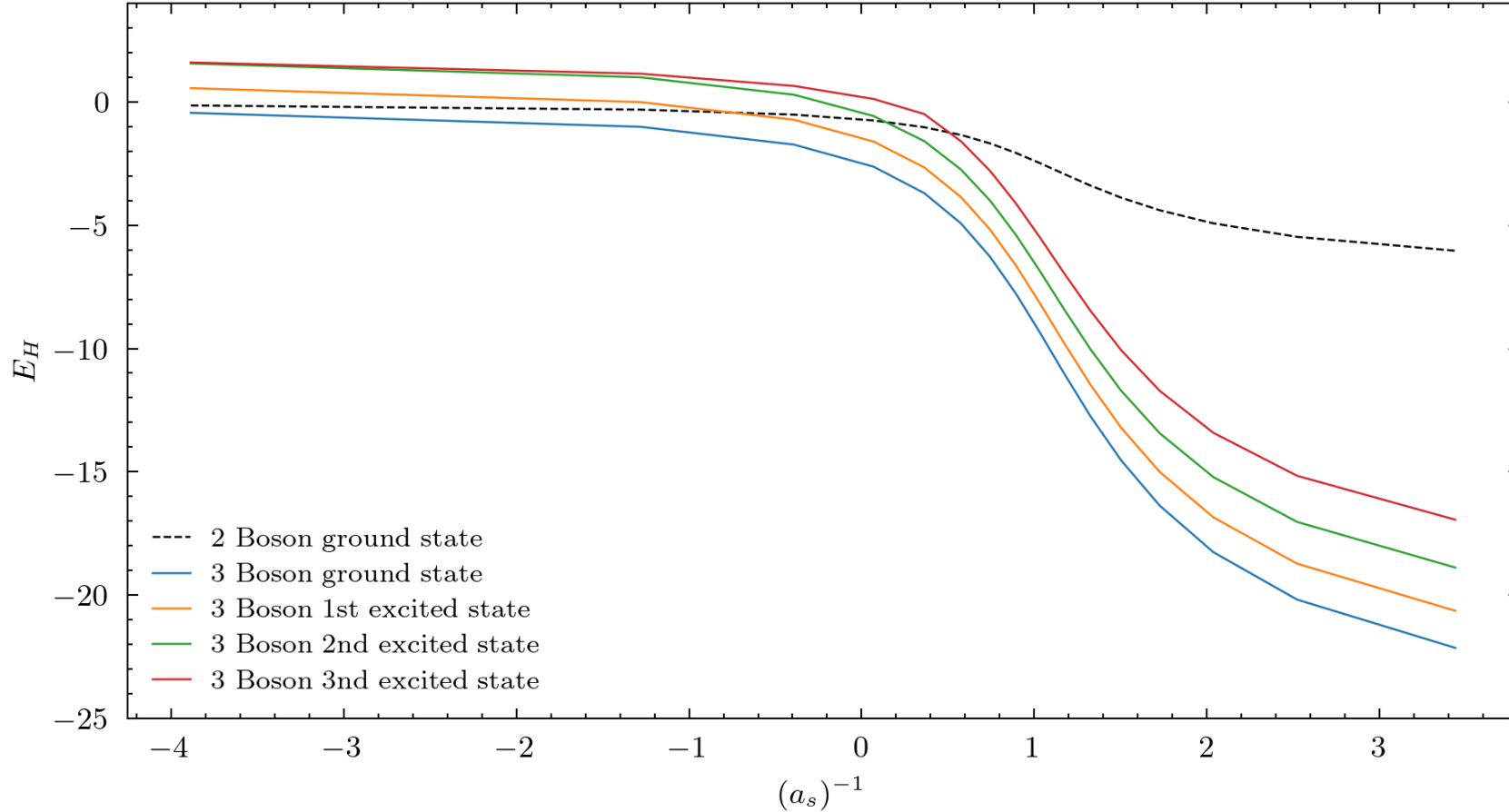


Figure 6: 2 Boson / 3 Boson Comparison

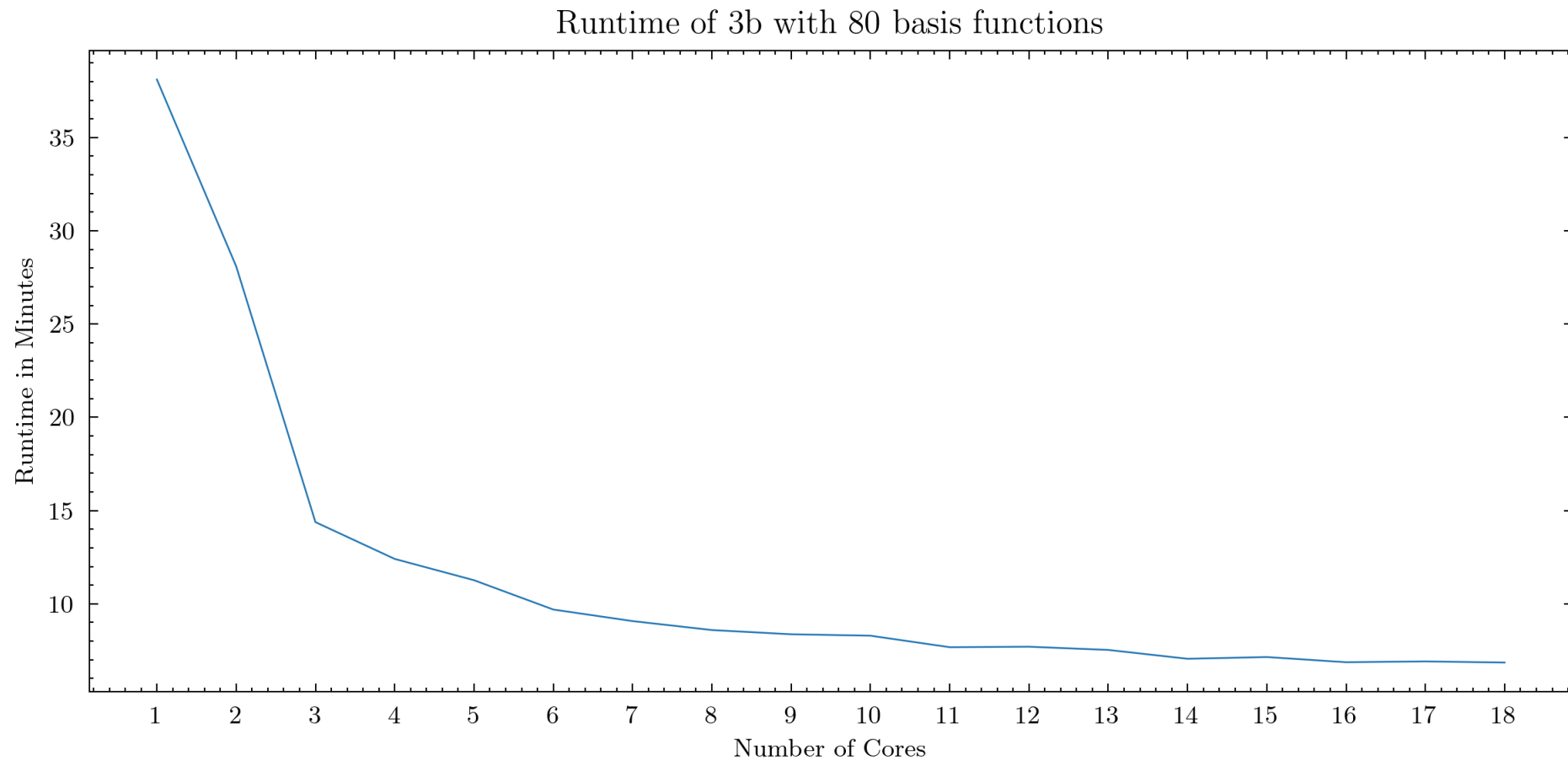


Figure 7: 2 Boson / 3 Boson Comparison

Computations

- CIR for more than 2 particles
- Efimov-Trimers
- n -body resonances
- Fermionization of Bosons in (quasi) 1d confinement

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- Convergence accelerator
- Arbitrary ext. potential
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Further Developments

- Angular interaction potentials
- DFT
- TDSE

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